

Project 3

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Requirements for submitting the assignment

- Email address : liujoy@seu.edu.cn
- Format of **Email subject** and **File name**:

StudentNumber-Name-Project-X

example : *71121000-张三-Project-1.zip*

- **Including :**

Submit only necessary code files such as **java/xml** etc. Other files that you think are important can also be included (such as README). The remaining project files do not need to be submitted.

Please limit the size of your code package.

① *report_A3.doc / report_A3.docx / report_A3.pdf*

② *code.zip*

Deadline: Beijing time, 11.20, 23:59:59

Project 3

I. Tutorial with Project Examples

II. Assignment

Tutorial with Project Examples

[Download JDK17 for Microsoft Windows](#)

1. Preparation Work

- **Install JDK17**  `jdk-17.0.17_windows-x64_bin.exe`
- **Change path (default path contains spaces, not recommended)**



将 Java(TM) SE Development Kit 17.0.17 (64-bit) 安装到:
C:\Program Files\Java\jdk-17\

更改...

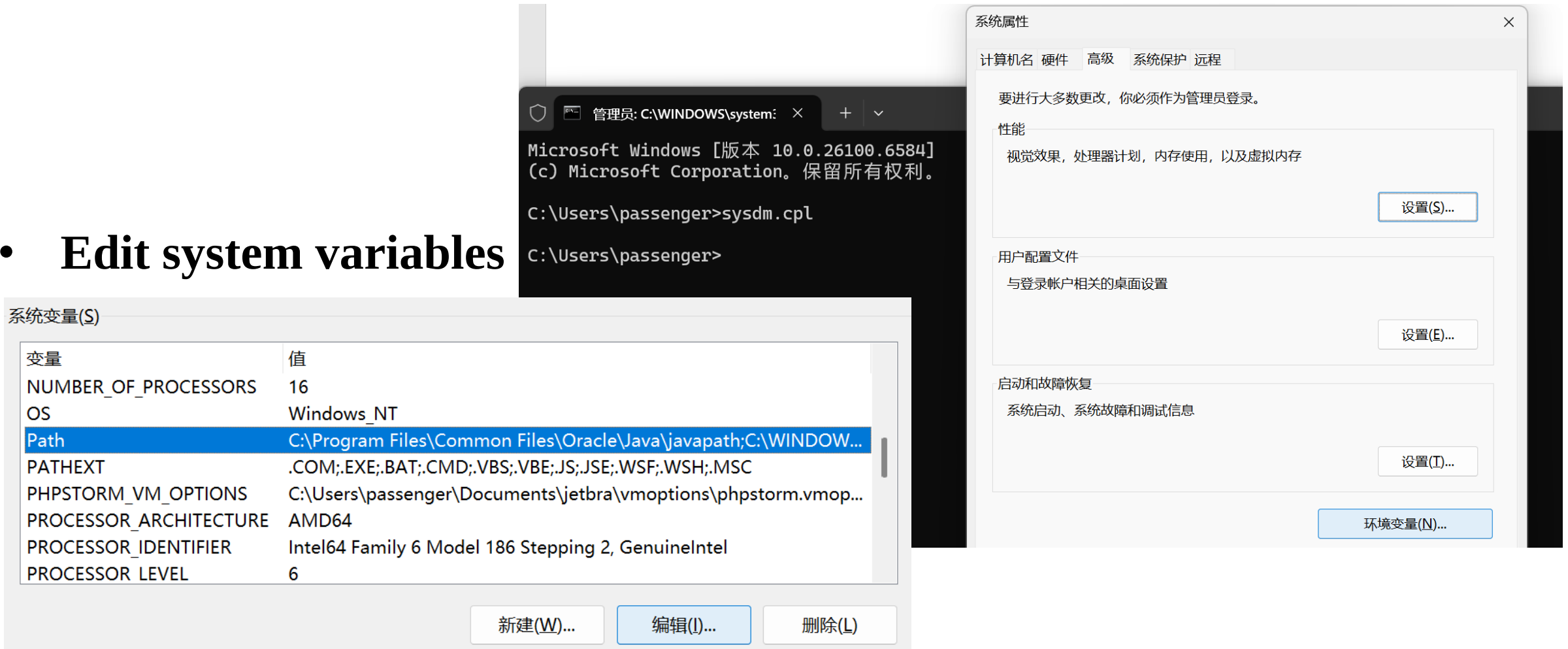
- **Installation completed**



Tutorial with Project Examples

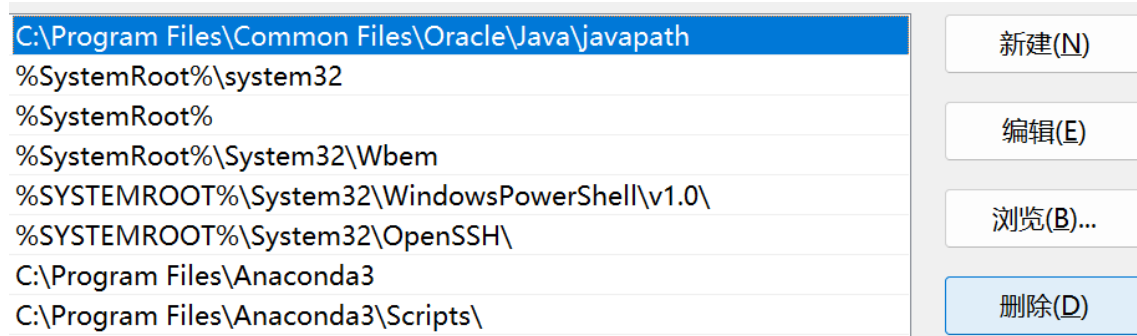
- Press Win+R, enter sysdm.cpl → Advanced → Environment Variables

- Edit system variables

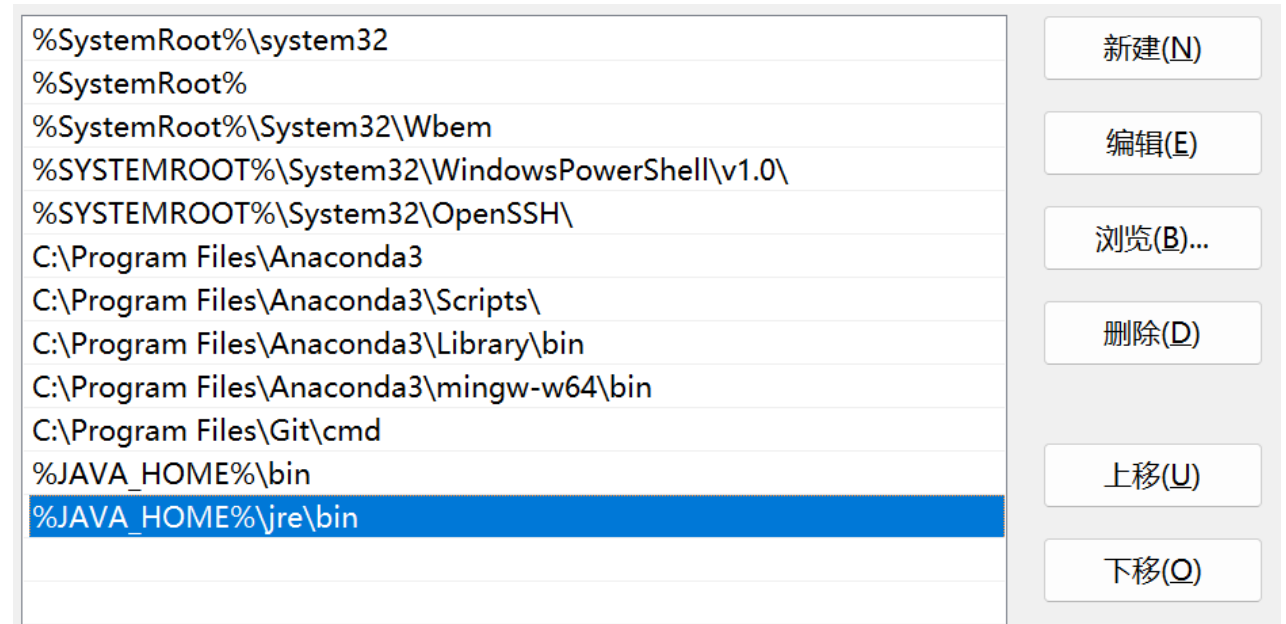


Tutorial with Project Examples

- **Delete**

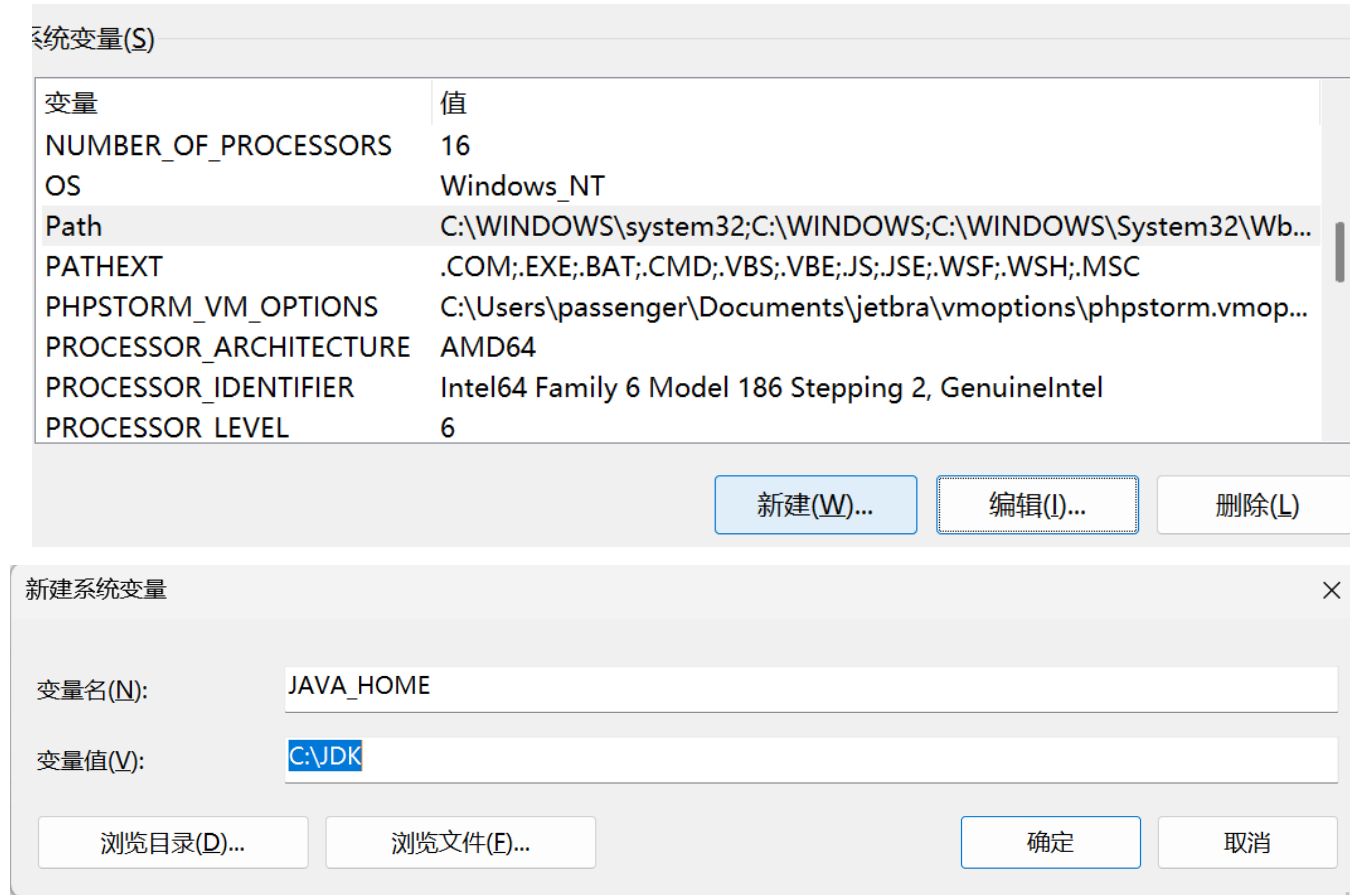


- **Add two variables in Path %JAVA_HOME%\bin and %JAVA_HOME%\jre\bin**



Tutorial with Project Examples

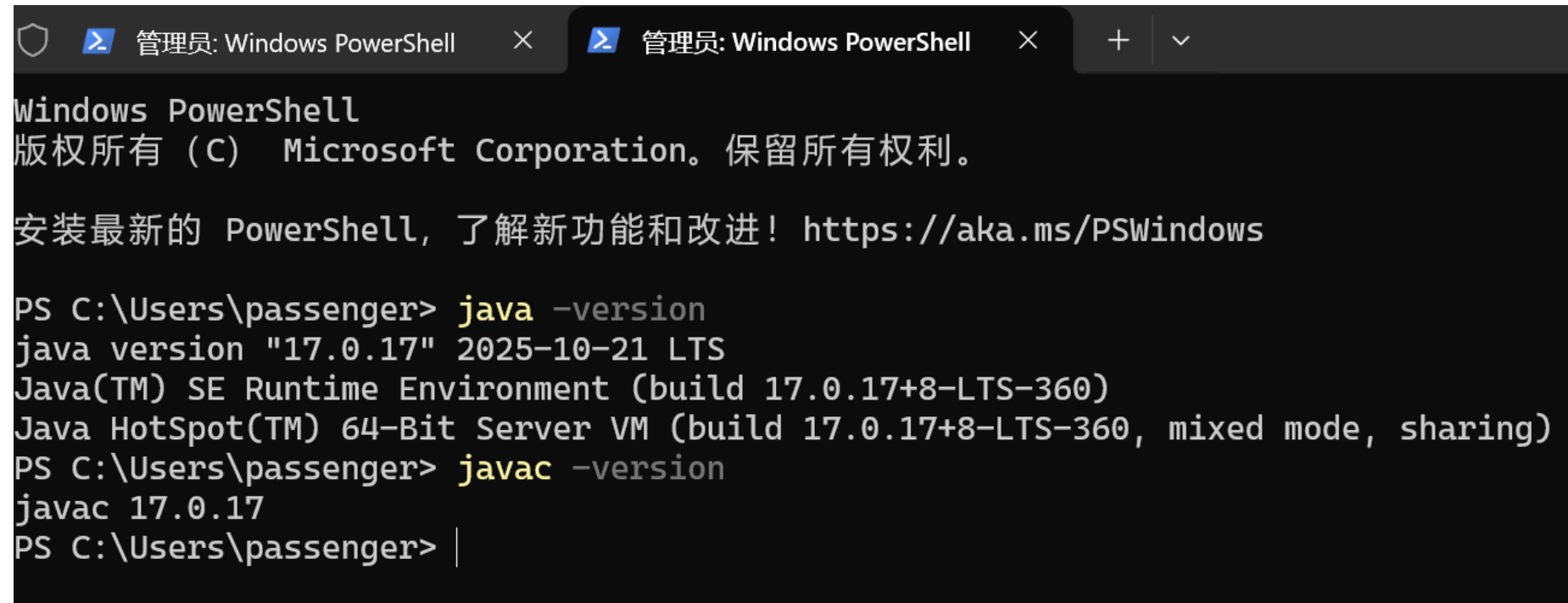
- Add the JAVA_SOME variable with the value of the jdk installation address



- Confirm all the way until all the windows are closed

Tutorial with Project Examples

- **Verify installation**
- **Press Win + R, enter cmd**
- **java -version**
- **javac -version**



```
Windows PowerShell
版权所有 (C) Microsoft Corporation。保留所有权利。

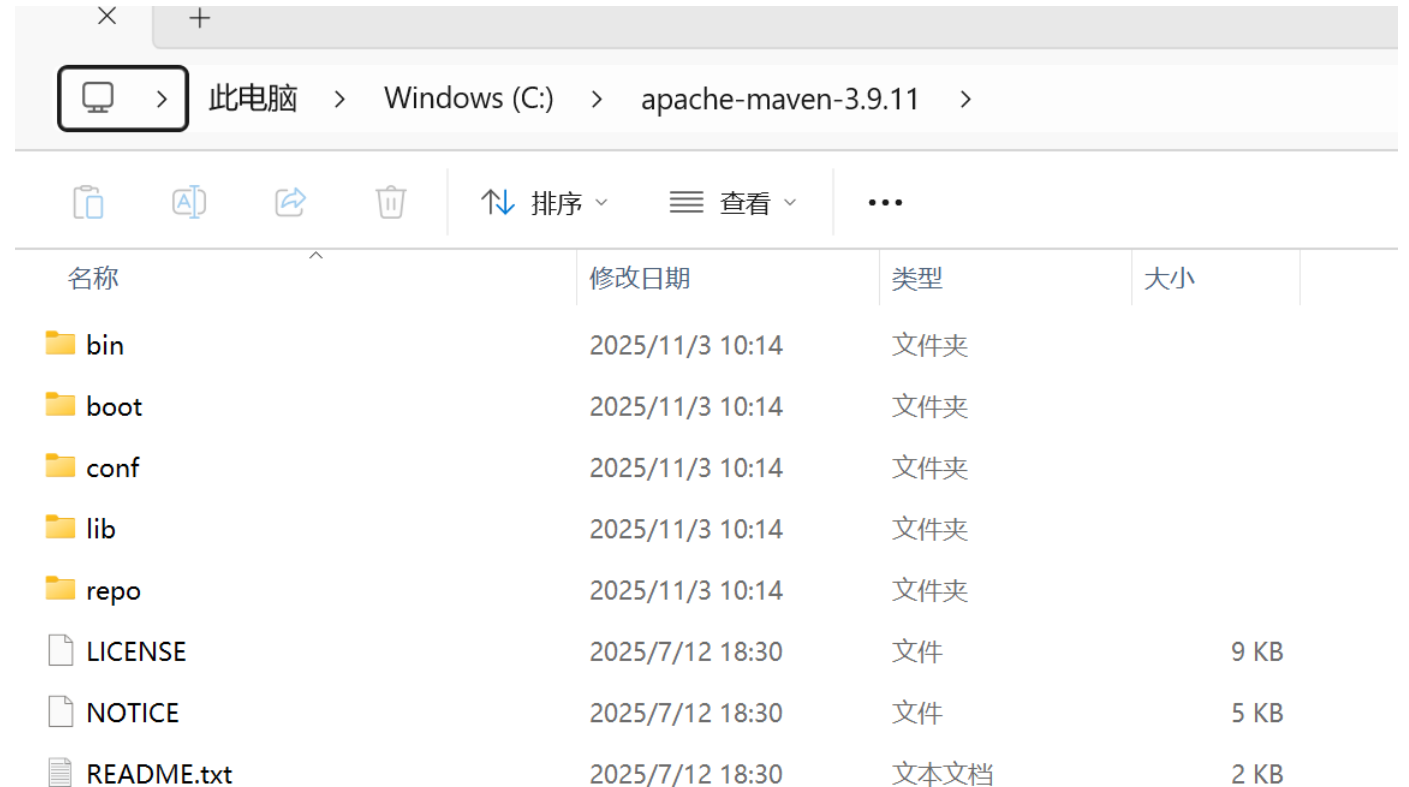
安装最新的 PowerShell, 了解新功能和改进! https://aka.ms/PSWindows

PS C:\Users\passenger> java -version
java version "17.0.17" 2025-10-21 LTS
Java(TM) SE Runtime Environment (build 17.0.17+8-LTS-360)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.17+8-LTS-360, mixed mode, sharing)
PS C:\Users\passenger> javac -version
javac 17.0.17
PS C:\Users\passenger> |
```

Tutorial with Project Examples

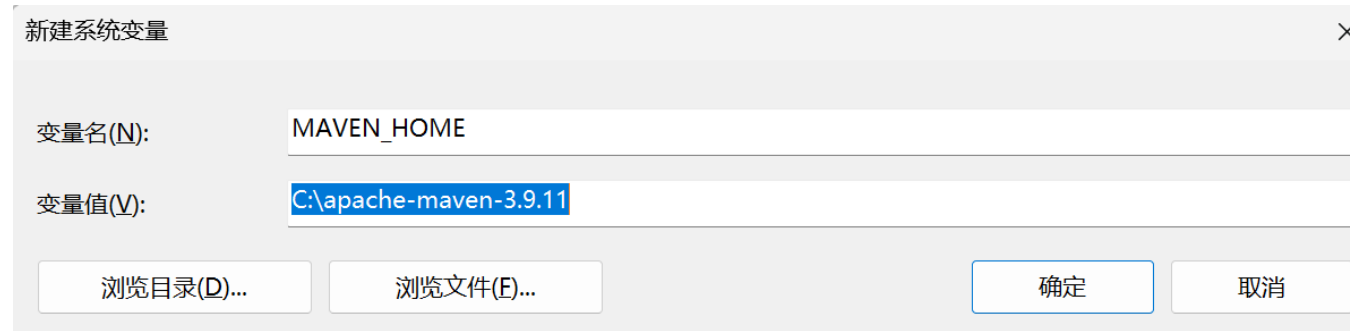
[Download Apache Maven – Maven](#)

- **Extract the downloaded Maven compressed file**
- **Enter the folder and create a repo folder to store Maven local repository dependencies**

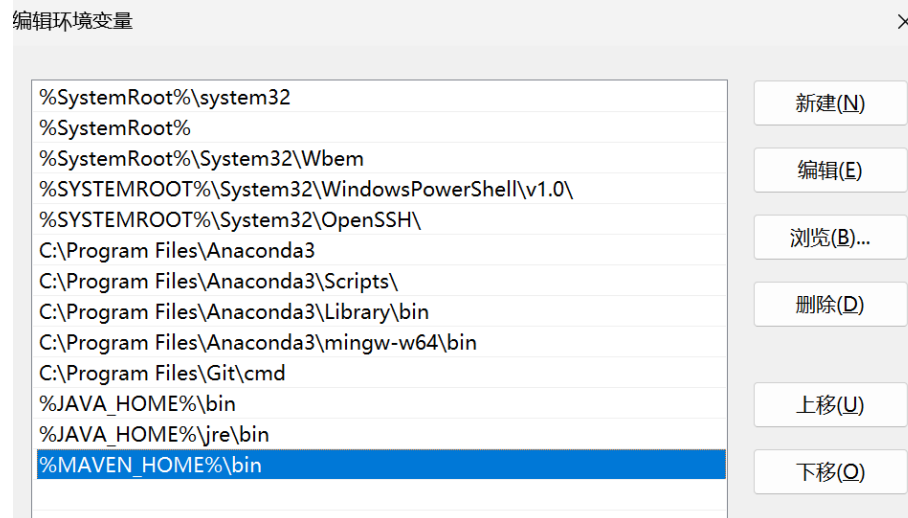


Tutorial with Project Examples

- Create a new system variable **MAVEN_SOME** with the value of the Maven installation address

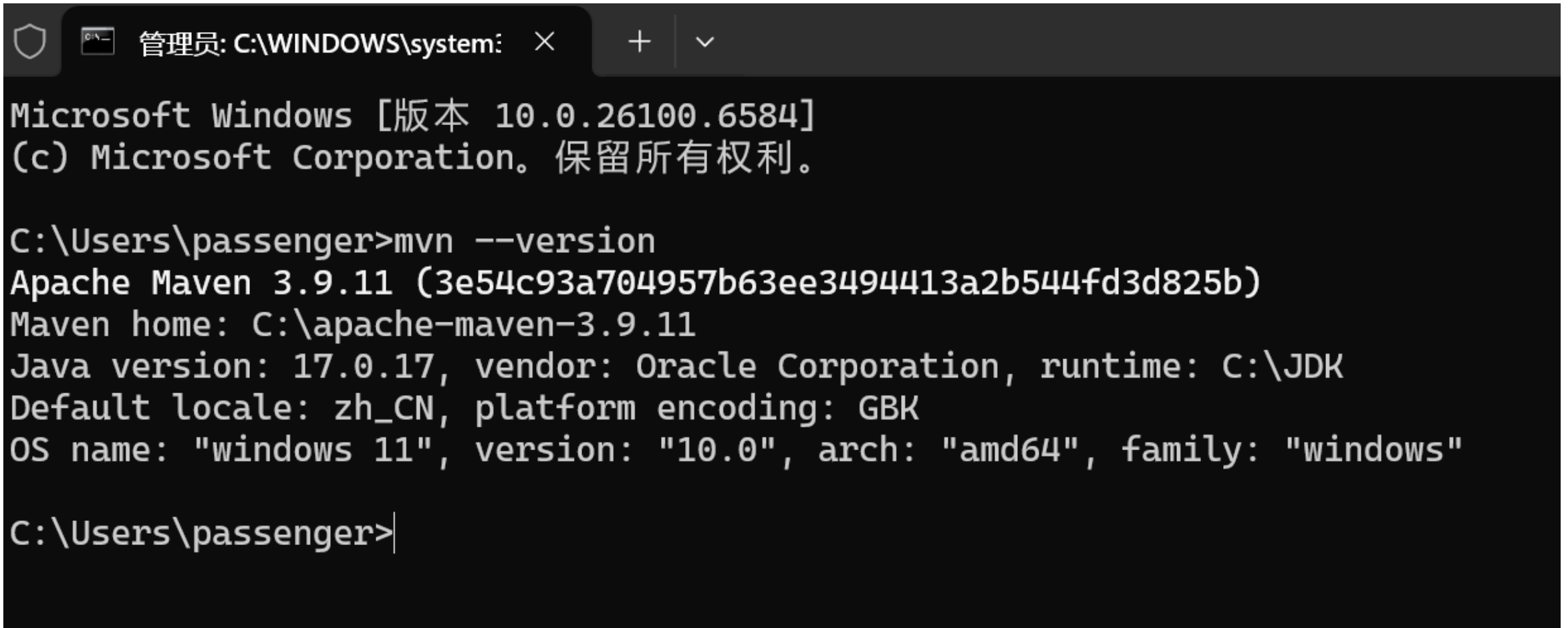


- Create a new environment variable **%MAVEN_HOME%\bin**



Tutorial with Project Examples

- **Win + R -> CMD , enter mvn --version (View Maven version)**



```
Microsoft Windows [版本 10.0.26100.6584]  
(c) Microsoft Corporation。保留所有权利。  
  
C:\Users\passenger>mvn --version  
Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)  
Maven home: C:\apache-maven-3.9.11  
Java version: 17.0.17, vendor: Oracle Corporation, runtime: C:\JDK  
Default locale: zh_CN, platform encoding: GBK  
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"  
  
C:\Users\passenger>
```

Tutorial with Project Examples

- Go to the config folder and edit the settings.xml file
- Specify repo address

```
<settings xmlns="http://maven.apache.org/SETTINGS/1.2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/SETTINGS/1.2.0 https://maven.apache.org/SETTINGS/1.2.0.xsd">
  <!-- localRepository
    | The path to the local repository maven will use to store artifacts.
    |
    | Default: ${user.home}/.m2/repository
  <localRepository>/path/to/local/repo</localRepository>
  -->
  <localRepository>C:\apache-maven-3.9.11\repo</localRepository>
```

- Configure mirrors

```
<!-- 阿里云镜像 -->
<mirror>
  <id>aliyun-maven</id>
  <mirrorOf>central</mirrorOf>
  <url>https://maven.aliyun.com/repository/public</url>
  <blocked>>false</blocked>
</mirror>
```

- Configure JDK version

The file sent to everyone is an edited version that does not require any changes except for the repo address

```
<!-- JDK17 -->
<profile>
  <id>jdk-17</id>
  <activation>
    <activeByDefault>>true</activeByDefault>
    <jdk>17</jdk>
  </activation>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <maven.compiler.compilerVersion>17</maven.compiler.compilerVersion>
  </properties>
</profile>
```

Tutorial with Project Examples

- Check if the configuration is successful
- Win + R -> CMD , enter mvn help:system (Print Maven information)

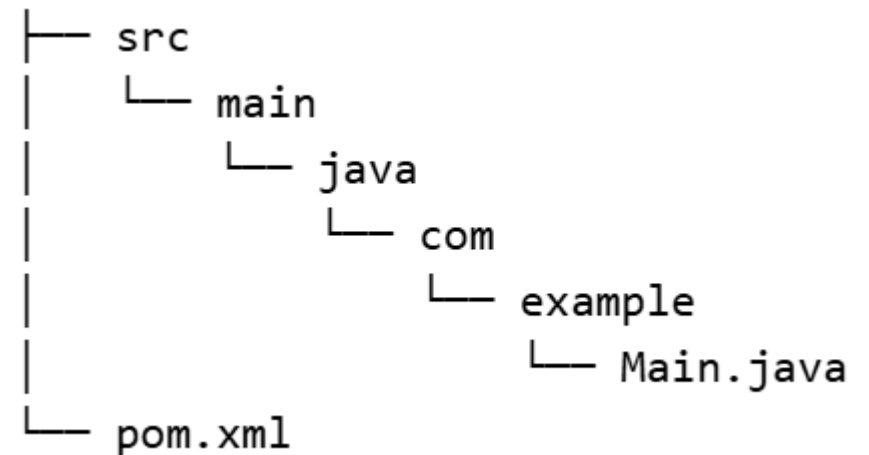
```
管理员: C:\WINDOWS\system32\cmd.exe
SYSTEMROOT=C:\WINDOWS
RUBYMINE_VM_OPTIONS=C:\Users\passenger\Documents\jetbra\vmoptions\rubymine.vmoptions
JETBRAINS_CLIENT_VM_OPTIONS=C:\Users\passenger\Documents\jetbra\vmoptions\jetbrains_client.vmoptions
MAVEN_CMD_LINE_ARGS=help:system
OS=Windows_NT
COMPUTERNAME=GW
COMMONPROGRAMW6432=C:\Program Files\Common Files
COMMONPROGRAMFILES=C:\Program Files\Common Files
COMSPEC=C:\WINDOWS\system32\cmd.exe
JAVACMD=C:\JDK\bin\java.exe
MAVEN_PROJECTBASEDIR=C:\Users\passenger
PROCESSOR_REVISION=ba02
CLASSWORLDS_LAUNCHER=org.codehaus.plexus.classworlds.launcher.Launcher
TEMP=C:\Users\PASSEN~1\AppData\Local\Temp
WEBIDE_VM_OPTIONS=C:\Users\passenger\Documents\jetbra\vmoptions\webide.vmoptions
HOMEDRIVE=C:
USERPROFILE=C:\Users\passenger
TMP=C:\Users\PASSEN~1\AppData\Local\Temp
NUMBER_OF_PROCESSORS=16
PHPSTORM_VM_OPTIONS=C:\Users\passenger\Documents\jetbra\vmoptions\phpstorm.vmoptions
JVMCONFIG=\.mvn\jvm.config

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 15.892 s
[INFO] Finished at: 2025-11-03T10:26:33+08:00
[INFO] -----
C:\Users\passenger>
```

Tutorial with Project Examples

- Create a new folder named LabsAccess JDBC as the project address
- Enter the folder, enter cmd in the address bar, and open the command line
- Enter at the command line:
- `mvn archetype:generate -DgroupId=com.example -DartifactId=access-jdbc-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false`

```
管理: C:\Windows\System32
7-4.jar (7.4 MB at 925 kB/s)
Downloaded from aliyun-maven: https://maven.aliyun.com/repository/public/com/ibm/icu/icu4j/77.1/icu4j-77
1.6 MB/s)
[INFO] Generating project in Batch mode
Downloading from aliyun-maven: https://maven.aliyun.com/repository/public/archetype-catalog.xml
[WARNING] failed to download from remoteorg.eclipse.aether.transfer.MetadataNotFoundException: Could not
archetype-catalog.xml in aliyun-maven (https://maven.aliyun.com/repository/public)
[WARNING] No archetype found in remote catalog. Defaulting to internal catalog
Downloading from aliyun-maven: https://maven.aliyun.com/repository/public/org/apache/maven/archetypes/m
ickstart/1.0/maven-archetype-quickstart-1.0.jar
Downloaded from aliyun-maven: https://maven.aliyun.com/repository/public/org/apache/maven/archetypes/mav
ckstart/1.0/maven-archetype-quickstart-1.0.jar (4.3 kB at 6.7 kB/s)
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickst
[INFO] -----
[INFO] Parameter: basedir, Value: C:\LabsAccessJDBC
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: access-jdbc-demo
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\LabsAccessJDBC\access-jdbc-demo
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 25.882 s
[INFO] Finished at: 2025-11-03T10:39:16+08:00
[INFO] -----
```



Tutorial with Project Examples

- Edit pom.xml

```
<dependencies>
  <!-- UCanAccess JDBC Driver for Access -->
  <dependency>
    <groupId>net.sf.ucanaccess</groupId>
    <artifactId>ucanaccess</artifactId>
    <version>5.0.1</version>
  </dependency>
```

Tutorial with Project Examples

- Operate the database by writing Java code

「被至少2个不同水手预订过」的船只名称		
bname		
tiger		
rose		
米		

```
SELECT bname
FROM Boats
WHERE bid IN (
    SELECT bid
    FROM (
        SELECT DISTINCT R.sid, R.bid
        FROM Reserves AS R
    ) AS DistinctReserves
GROUP BY bid
HAVING COUNT(*) >= 2
);
```

```
public class Main {
    private static final String DB_PATH = "C:\\Users\\passenger\\Documents\\project2.accdb";
    private static final String DB_URL = "jdbc:ucanaccess://" + DB_PATH;

    Run | Debug
    public static void main(String[] args) {
        try {
            // UCanAccess 驱动 (Maven 已引入, 无需手动加载, 但保留更清晰)
            Class.forName(className: "net.ucanaccess.jdbc.UcanaccessDriver");

            Connection conn = DriverManager.getConnection(DB_URL);
            System.out.println(x: "✅ 成功连接 Access 数据库!");

            // 执行 CRUD
            selectTest(conn);
            conn.close();
            System.out.println(x: "🔌 连接已关闭。");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    private static void selectTest(Connection conn) throws SQLException {
        String sql = "SELECT bname\r\n" + //
            "FROM Boats\r\n" + //
            "WHERE bid IN (\r\n" + //
            "    SELECT bid\r\n" + //
            "    FROM (\r\n" + //
            "        SELECT DISTINCT R.sid, R.bid\r\n" + //
            "        FROM Reserves AS R\r\n" + //
            "    ) AS DistinctReserves\r\n" + //
            "    GROUP BY bid\r\n" + //
            "    HAVING COUNT(*) >= 2\r\n" + //
            "    );\r\n" + //
            "    ";
        try (Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(sql)) {
            System.out.println(x: "\n--- 查询结果 ---");
            while (rs.next()) {
                System.out.printf(format: "姓名: %s \n",
                    rs.getString(columnLabel: "bname"));
            }
        }
    }
}
```

```
PS C:\LabsAccessJDBC\access-jdbc-demo> c:;
rt=dt_socket,server=n,suspend=y,address=loc
'com.example.Main'
? 成功连接 Access 数据库!

--- 查询结果 ---
姓名: tiger
姓名: rose
? 连接已关闭。
PS C:\LabsAccessJDBC\access-jdbc-demo>
```

Project 3

I. Tutorial with Project Examples

II. Assignment

Assignment

1. Assignment Content

1. Connect to the universities database using Java
2. Implement adding, deleting, updating, and querying database tables using Java

Assignment

1. Assignment Content

3. Questions

This assignment requires everyone to exercise their subjective initiative:

1. Based on existing tables in the database, create a new table and design the fields and primary keys for the created table. The report needs to explain the reasons for establishing this table.
2. Delete the newly created table.
3. Insert a row of record in the student table.
4. Delete the newly inserted record.
5. Update a record in the student table.
6. Restore the modified records to their original state.
7. Execute all the SQL statements in Project 2.

Assignment

1. Assignment Content

4. Schema and Sample Data

- The schema of the database is provided below (keys are red, field types are omitted):
 - student(**sid**, sname, sex, age, year, gpa)
 - dept(**dname**, numphds)
 - prof(**pname**, dname)
 - course(**cno**, cname, **dname**)
 - major(**dname**, **sid**)
 - section(**dname**, **cno**, **sectno**, pname)
 - enroll(**sid**, grade, **dname**, **cno**, **sectno**)

Assignment

1. Assignment Content

5. Data Files

- Download the following data files from WeChat Group (The imports you made in Project 1).
 - **course.txt**
 - **dept.txt**
 - **enroll.txt**
 - **major.txt**
 - **prof.txt**
 - **section.txt**
 - **student.txt**

It is recommended to use the original **student.txt** for better communication and comparison of correctness among classmates.

Assignment

2. A Few Points to Note

- Pay attention to the address of the database connection, a single slash is an escape character.
- This is an **individual assignment** – no group submissions are allowed.
- Hand in **a report** which outlines your answers and the process by which you arrived at them.